Worksheet

1. Compute $< 3, 5, -2 > -4 < -1, 0, 3 >$.

2. Find $||2\mathbf{i} - 3\mathbf{j} + \mathbf{k}||$.

3. Find a unit vector that points in the same direction as $< 5, -1, 4 >$.

4. Find the angle between $-\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ and $5\mathbf{i} - \mathbf{j}$.

5. Are $-\mathbf{i} + 2\mathbf{j} + 4\mathbf{k}$ and $5\mathbf{i} - \mathbf{j}$ perpendicular to each other?

6. Are $< 2, 0, -3 >$ and $< 1, -2, 1 >$ orthogonal to each other?

7. Find $< 5, -1, 4 > x < 3, 0, 1 >$

8. Find a unit vector that is orthogonal to $< 5, -1, 4 >$ and $< 3, 0, 1 >$.